

March 24, 2026

Editor-in-Chief
IEEE Transactions on Computer-Aided Design of
Integrated Circuits and Systems (TCAD)

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled “**Building VHDL Processors with Claude Code: From 16-Bit to 64-Bit FPU**” for consideration as a journal paper in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems.

This manuscript is an extended version of our conference paper presented at the ACM/IEEE Workshop on Machine Learning for CAD (MLCAD 2026). The journal version contains more than 30% new content beyond the conference paper, including the following substantial additions:

1. **Comprehensive Comparison with Prior Work (Table III, Section V):** A new dedicated section presents a detailed comparison with five prior approaches to AI-assisted HDL generation (DAVE, Chip-Chat, HDLGen-ChatGPT, ResBench, and this work) across ten quantitative metrics, accompanied by approximately one page of narrative analysis comparing methodologies, qualitative differences, and trade-offs.
2. **Timing Analysis and Per-Variant Resource Breakdown (Section IV.E, Table IV):** A new section provides critical path analysis, maximum frequency estimation ($F_{\max} \approx 108.7$ MHz), Verilog vs. VHDL synthesis comparison, and a detailed per-variant resource breakdown (LUTs, flip-flops, DSP48E1, BRAM) for all three processor variants in both languages.
3. **Extended FPU Verification Coverage (Section IV.C, Table III):** A new subsection presents an IEEE 754 corner-case coverage matrix documenting which special values (denormals, $\pm\infty$, qNaN, sNaN, ± 0) are tested for each of the 28 FPU operations via simulation and formal verification, along with the test vector selection methodology and coverage analysis.
4. **Toward Unbounded Verification (Section IV.F):** A new section discusses the limitations of bounded model checking and presents strategies for extending verification toward unbounded proofs, including k-induction, abstract interpretation, assume-guarantee decomposition, and IC3/PDR algorithms, with a concrete four-phase roadmap.
5. **Extended Related Work (Section I):** The introduction has been expanded with a survey of five additional recent works in AI-assisted hardware design: RTLCoder (Liu et al., 2024), ChipNeMo (NVIDIA, 2024), VeriGen (Thakur et al., 2024), RTLLM (Lu et al., 2024), and ChipGPT (Chang et al., 2023).
6. **Reproducibility Section (Section II.E):** A new subsection details the skill file structure (1,286 lines across four files), the three-phase generation workflow, build automation infrastructure, and open-source availability.

The new content exceeds 30% of the total manuscript, expanding the conference paper from approximately 10 pages to 15 pages in IEEE journal format. All original content from the conference paper has been preserved and, where appropriate, strengthened with cross-references to the new material.

The manuscript has not been submitted elsewhere and is not under consideration by any other journal. All authors have approved the manuscript and agree to its submission.

We believe this work makes a significant contribution to the field of computer-aided design by demonstrating that domain-specific context engineering can enable AI-assisted generation of hardware designs far exceeding the complexity of any prior work — including a 64-bit RISC-V processor with a full IEEE 754 double-precision floating-point unit, verified through both simulation-based TDD and formal methods.

We respectfully request that this manuscript be considered for publication in IEEE TCAD.

Sincerely,

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